

William Randolph Oak

SENIOR SOFTWARE ARCHITECT | AI-FIRST ENGINEERING | DISTRIBUTED SYSTEMS

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EXECUTIVE PROFILE

Senior software architect and engineer with 30+ years of experience turning ambiguous, high-risk technical problems into maintainable systems. Combines deep production experience in distributed, real-time, telecom, web, Linux, and legacy environments with an AI-first engineering practice. Brings the judgment to distinguish useful innovation from avoidable complexity, establish sound foundations, and communicate technical tradeoffs to both engineering teams and organizational leadership.

ARCHITECTURE UNDER AMBIGUITY	PRACTICAL AI ADOPTION	DELIVERY FOUNDATIONS
Frames the problem, evaluates tradeoffs, and designs for resilience, scale, operability, and change.	Uses AI for research, validation, debugging, testing, review, and documentation with experienced human oversight.	Establishes standards, CI/CD direction, documentation, testability, and team practices where little exists.

PROFESSIONAL EXPERIENCE

- AI Integration Advisor / Systems Architect - Harding Instruments

2024-2025

 - Led the design and implementation of a distributed communications platform from a zero-development starting point, translating operational requirements into a scalable technical foundation.
 - Defined a Kubernetes-based architecture for horizontal scaling, fault tolerance, self-healing, and offline operation, including replicated and distributed storage.
 - Established a single-source-of-truth API with real-time WebSocket propagation and built TypeScript services spanning SIP, ESL, Hostport, and legacy protocols.
 - Used AI continuously across architecture validation, cross-domain research, debugging, design review, and technical decision-making during a 21-month engagement.
 - Changed the platform direction to a unified TypeScript ecosystem after evaluating architectural and delivery tradeoffs; reverse-engineered the legacy system to preserve compatibility.
 - Defined engineering standards, documentation strategy, development environment, and CI/CD requirements needed to move from prototype conditions toward sustainable delivery.
- Senior Software Engineer - Veritree

2023

 - Designed a STRAPI configuration architecture that enabled controlled client content updates; practiced TDD with PHPUnit and contributed to GitHub-to-AWS delivery workflows.
- Senior Software Engineer - Zau

2019-2023

 - Developed and maintained SaaS platform capabilities, mentored developers, supported Agile delivery, and contributed to CI/CD and test-driven development practices.
- Senior Software Engineer - XenCALL

2016-2019

 - Built real-time telecom capabilities using PHP, JavaScript, MySQL, Memcached, WebSockets, and Asterisk; led delivery cycles and mentored developers.

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Senior Software Architect | AI-First Engineering | Distributed Systems

SELECTED ENGINEERING CAPABILITIES

SYSTEMS ARCHITECTURE	Distributed and fault-tolerant systems; cluster topology; horizontal scaling; offline-capable design; architecture planning under ambiguity.
REAL-TIME COMMUNICATIONS	SIP, FreeSWITCH, Asterisk, ESL, Hostport, WebSockets, event-driven coordination, and protocol translation.
BACKEND AND INTEGRATION	TypeScript/Node.js services, PHP platforms, REST APIs, single-source-of-truth design, data flow, versioned endpoints, and legacy integration.
AI-FIRST ENGINEERING	AI-assisted research, architecture validation, debugging, code review, test generation, documentation, and rapid acquisition of unfamiliar technologies.
ENGINEERING LEADERSHIP	Technical tradeoff analysis, standards, CI/CD direction, documentation strategy, mentoring, Agile delivery, and communication across stakeholder groups.
INFRASTRUCTURE AND OPERATIONS	Linux, Kubernetes, GlusterFS, distributed storage, Apache, DNS, DHCP, AWS delivery workflows, production support, and automation.

EARLIER EXPERIENCE

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| Programmer & Network Consultant - Northshore Driving School | 2000-2016 |
| • Designed and supported scheduling, networking, hosting, and automation solutions in a long-term consulting relationship. | |
| Programmer & Network Specialist - Website-in-a-Box | 2007-2016 |
| • Built and maintained CMS, web-hosting, and infrastructure capabilities across client environments. | |
| Team Lead - Mission:Impossible - UFCW Local 1518 | 2011-2012 |
| • Led project delivery and coordination in a union environment. | |

TECHNOLOGY

Languages	TypeScript, JavaScript, PHP, Bash, Perl, SQL, Python
Platforms	Node.js, React, Linux, Kubernetes, Apache, AWS workflows
Data & protocols	PostgreSQL and compatible systems, MySQL, Memcached, REST, WebSockets, SIP, ESL
Engineering	TDD, PHPUnit, CI/CD, GitHub, API documentation, distributed storage

EDUCATION & CONTINUOUS DEVELOPMENT

Machine Learning - Stanford University (Coursera)
Completed March 12, 2019 | Grade: 97.25% | Instructor: Andrew Ng
coursera.org/account/accomplishments/verify/DEMPANGXTAH

Current portfolio: Active repositories demonstrate modern TypeScript API architecture, automated testing, versioned routing, API documentation, and AI-assisted development practices.

References available upon request.